Anush Reddy Mudiganti

Pleasanton, CA • Foothill High School, Class of 2027 • GPA: 4.4 • SAT: 1450

EDUCATION

Foothill High School | Pleasanton, CA | Class of 2027
GPA: 4.4 | Honor Roll | California Scholarship Federation (CSF)

Current Courses:
AP Computer Science A - **5** • AP Calculus BC - **5** • AP U.S. History - **5** • AP Music Theory - **3** • Honors Principles of Engineering

Upcoming (2025–26):
AP Government • AP Macroeconomics • AP Physics C • AP Statistics • Multivariable Calculus • Honors Computer Integrated Manufacturing • Honors Aerospace Engineering

TECHNICAL SKILLS

Languages: Java, Python, C++, C#, JavaScript, TypeScript, HTML, CSS, Lua
Frameworks & Tools: React, Astro, Tailwind, Node.js, Supabase, Google Firebase, Git, WSL
Hardware: Custom PCB design, Raspberry Pi, 3D printer hardware, microcontrollers

PROJECTS & COMPETITIONS

NASA Space Apps Challenge 2025
- Built a website using NASA weather data to analyze and visualize atmospheric patterns across geographic regions.

Alameda County Science & Engineering Fair 2025
- Researched the connection between conducting gesture and the circle of fifths to help conductors assist ensemble sections in staying in tune.
- Won 3rd Place.

Noisele — Daily Wordle-Inspired Music Game WIP
- Web game where players identify musical noise/tones daily; built with HTML, CSS, JavaScript, React, Astro, and Supabase.

3D Orbit Simulation with Procedural Planet Generation WIP
- Real-time simulation with procedurally generated planets using multiple noise algorithms; built in C# and Python.

Light-Painting Robot 2025
- Robot that takes image input and replicates it as long-exposure light photographs. Coded entirely in C++ with custom hardware. Repurposed 3D printer components to minimize cost.

Mini Music Player (Raspberry Pi) WIP
- Portable music player with a custom PCB, LCD screen, headphone jack, and SD card reader. Software written in Python, HTML, CSS, and JavaScript.

Custom Macroboard 2026
- Designed a custom PCB and using modified QMK firmware

Online Jeopardy Game with Lobby System WIP
- Multiplayer Jeopardy platform with real-time lobbies using HTML, CSS, JavaScript, SQL, and Google Firebase.

Task App — Homework Tracker WIP
- Full-stack web app to track homework assignments with custom authentication and Supabase backend.

Image-to-SVG / SVG-to-GCode Converter
- Website that converts raster images to SVGs and SVGs to GCode for CNC/plotter use.

2D Orbit Simulation 2024
- Newtonian physics-based 2D orbital mechanics simulator written in C++.

Grocery Reminder App WIP
- App that reminds users to buy groceries at a scheduled future time.

ACTIVITIES & LEADERSHIP

Math Club Foothill High School

Physics Club Foothill High School

VOLUNTEERING

Student Volunteer, California Scholarship Federation (CSF) 2024 – Present | ~2 hrs/wk | ~144 hrs total

- Collected and donated items for nonprofit organizations through CSF fundraising drives.
- Created handmade cards for senior citizens and cancer patients.
- Distributed T-shirts to PPIE race participants.

Volunteer, Cycle of Hope — Habitat for Humanity 2024 – Present

- Race Day (7 hrs): Handed out bracelets and food/water to racers, assembled goody bags, set up aid stations along the course, and helped unload supplies.
- Pre-Race Setup (5 hrs): Registered participants, distributed race numbers and event information, assembled goody bags, and loaded supplies onto trucks at a local bike shop.

FINE ARTS

Piano 13+ Years

- Numerous state and national awards in competitive performance.
- Piano Guild: consistent Superior scores for 10+ years.
- MTAC State participant.
- AP Music Theory — formal study of theory alongside performance.

ADDITIONAL INFORMATION

Independent Study

- Self-studying foundational physics over summer 2026 to prepare for AP Physics C, having skipped the standard Physics course.